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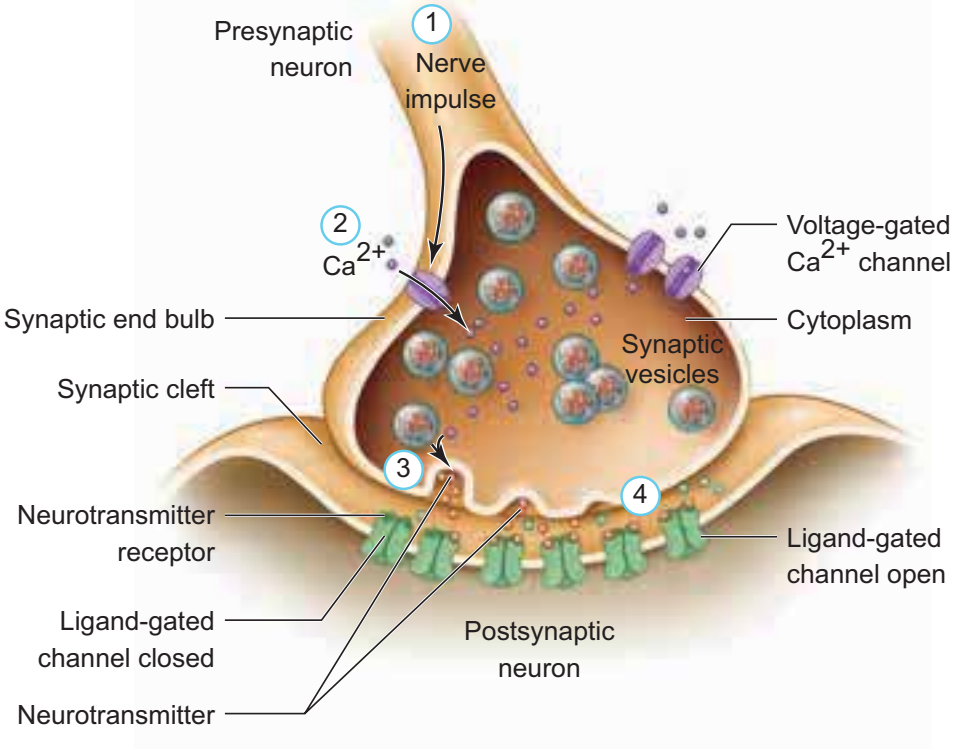

Figure 10.6 Synaptic Transmission

impulse is transmitted. The first neuron involved in the synapse forms the **pre-synaptic neuron** and the second neuron is **the post-synaptic neuron**. A small gap between the pre and postsynaptic membranes is called **Synaptic Cleft** that forms a structural gap and a functional bridge between neurons. The axon terminals contain synaptic vesicles filled with **neurotransmitters**. When an impulse [action potential] arrives at the axon terminals, it depolarizes the pre-synaptic membrane, opening the voltage gated calcium channels. Influx of calcium ions stimulates the synaptic vesicles towards the pre-synaptic membrane and fuses with it. In the neurilemma, the vesicles release their neurotransmitters into the synaptic cleft by **exocytosis.** The released neurotransmitters bind to their specific receptors on the post-synaptic membrane, responding to chemical signals. The entry of the ions can generate a new potential in the post-synaptic neuron, which may be either **excitatory** or **inhibitory**. Excitatory post-synaptic potential causes depolarization whereas inhibitory post-synaptic potential causes **hyperpolarization** of post-synaptic membrane (Figure 10.6).

Can you state why some areas of the brain and spinal cord are grey and some are white?

10.4 Central neural system (CNS)

The CNS includes the brain and the spinal cord, which are protected by the bones of the skull and vertebral column. During its embryonic development, CNS develops from the ectoderm.

10.4.1 Brain

The brain acts as the command and control system. It is the site of information processing. It is located in the cranial cavity